# Finite generation of pluricanonical rings

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# Outline of the talk

1. Introduction

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   - The main Theorem
   - Curves
   - Surfaces
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- $K_X$ denotes (a choice of) the canonical divisor i.e. a formal linear combination of codimension 1 subvarieties of $X$ given by the zeroes of a section $s \neq 0$ of $\omega_X$.
- We would like to use $\omega_X$ to understand the geometry of $X$.

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- If $\kappa(X) = \dim X$, we say that $X$ is of **general type**.
- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

# Finite generation

Today I would like to discuss a recent result of Birkar-Cascini-Hacon-M$^c$Kernan and Siu:

**Theorem**

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The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires $X$ to be of general type.

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- $g \geq 2$: Curves of general type.
  These form a $3g - 3$-dimensional family.
  $K$ is ample and $|3K|$ determines an embedding.
- $\mathfrak{R}(K)$ is then finitely generated (by Serre-Vanishing).

# Surfaces of general type

- If $S$ is a surface of general type and $K_S$ is nef (i.e. $K_S \cdot C \geq 0$ for all curves $C \subset S$), then we say that $S$ is a minimal surface. In this case $K_S$ is semiample.

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- Note that we may take $m = 5$, $\nu$ contracts rational curves $E$ with $E \cdot K_S = 0$ and $\bar{S} = \mathrm{Proj}\mathfrak{R}(K_S)$.

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- Then $\mathfrak{R}(K_{S_t})$ is finitely generated and so is $\mathfrak{R}(K_S)$.

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- We have that $mK_{S'} = f^*H$ where $m > 0$ is an integer and $H$ is an ample divisor on $C$.
- Therefore, $\mathfrak{R}(K_S)^{(m)} \cong \mathfrak{R}(K_{S'})^{(m)} \cong \mathfrak{R}(H)$ is finitely generated.

Reduction to log pairs of general type

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   - Reduction to log pairs of general type
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- The proof will in fact proceed by induction on the dimension. We will need to relate $K_X$ to it's restriction to a divisor $S$ (which typically belongs to $S \in |mK_X|$).
- This is achieved by the adjunction formula which has the form $(K_X + S)|_S = K_S + \Delta_S$ for some $\mathbb{Q}$-divisor $\Delta_S$.

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- $(X, \Delta)$ is KLT if each coefficient of $\Delta'$ is less than 1.

# A Theorem of Fujino-Mori

Mori and Fujino have shown that:

**Theorem**

*Let $(X,\Delta)$ be any KLT $\mathbb{Q}$-factorial pair with $\Delta \in \mathrm{Div}_{\mathbb{Q}}(X)$. Then there exists a* **big** *KLT $\mathbb{Q}$-factorial pair $(Y,\Gamma)$ with $\Gamma \in \mathrm{Div}_{\mathbb{Q}}(Y)$ such that*

$$\mathfrak{R}(K_X+\Delta)^{(m)} \cong \mathfrak{R}(K_Y+\Gamma)^{(m)}$$

*for any sufficiently divisible integer $m>0$.*

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Recall that $K_Y+\Gamma$ is big if $\mathrm{tr.deg.}_{\mathbb{C}}\mathfrak{R}(K_Y+\Gamma)^{(m)} = \dim Y$ or equivalently if $K_Y+\Gamma \sim_{\mathbb{Q}} A+B$ with $A$ ample and $B \geq 0$.

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It follows that to show the main theorem (i.e. that $\mathfrak{R}(K_X)$ is finitely generated), it suffices to prove the corresponding statement for big KLT log pairs.

Reduction to log pairs of general type

## The MMP with scaling

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- Note that if $K_X + \Delta$ is big, then there is an effective divisor $D \sim_{\mathbb{R}} K_X + \Delta$ and so for all $0 < \epsilon \ll 1$ the pair $(X, \Delta + \epsilon D)$ is KLT, $\Delta + \epsilon D$ is big and $K_X + \Delta + \epsilon D \sim_{\mathbb{R}} (1 + \epsilon)(K_X + \Delta)$.

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- We would like to find a finite sequence of KLT log pairs $(X_i,\Delta_i)$ with $\Delta_i$ big and birational maps $X \dashrightarrow X_1 \dashrightarrow X_2$ such that

$$\mathfrak{R}(K_{X_i}+\Delta_i) \cong \mathfrak{R}(K_{X_{i+1}}+\Delta_{i+1}),$$

and $K_{X_t}+\Delta_t$ is nef.

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- From now on we will consider $\mathbb{Q}$-factorial KLT pairs $(X, \Delta)$ such that $\Delta$ is big. We allow $\Delta \in \mathrm{Div}_{\mathbb{R}}(X)$.
- Note that if $K_X + \Delta$ is big, then there is an effective divisor $D \sim_{\mathbb{R}} K_X + \Delta$ and so for all $0 < \epsilon \ll 1$ the pair $(X, \Delta + \epsilon D)$ is KLT, $\Delta + \epsilon D$ is big and $K_X + \Delta + \epsilon D \sim_{\mathbb{R}} (1+\epsilon)(K_X + \Delta)$.
- We would like to find a finite sequence of KLT log pairs $(X_i, \Delta_i)$ with $\Delta_i$ big and birational maps $X \dashrightarrow X_1 \dashrightarrow X_2$ such that

$$\mathfrak{R}(K_{X_i} + \Delta_i) \cong \mathfrak{R}(K_{X_{i+1}} + \Delta_{i+1}),$$

  and $K_{X_t} + \Delta_t$ is nef.
- Then by the base point free theorem, $K_{X_t} + \Delta_t$ is semiample so that $\mathfrak{R}(K_{X_t} + \Delta_t)$ is finitely generated (if $\Delta \in \mathrm{Div}_{\mathbb{Q}}$).

# The MMP with scaling II

- Assume that $A$ is ample and $K_X + \Delta + A$ is nef.

Reduction to log pairs of general type

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- Let $\mathrm{cont}_R : X\to Z$ be the corresponding morphism which contracts a curve $C$ iff $[C]=R$.
- If $\mathrm{cont}_R$ is not birational, we STOP (we have a Mori-Fano fiber space and $\mathfrak{R}(K_X+\Delta)\cong\mathbb{C}$).

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- Instead we replace $(X, \Delta)$ by the corresponding flip $\phi : X \dashrightarrow X^+$ (assuming it exists).
- Let $\Delta^+ = \phi_* \Delta$. The flip is a codimension 2 surgery which replaces $K_X + \Delta$ negative curves by $K_{X^+} + \Delta^+$ positive curves.

# Flips

- More precisely, $f = \mathrm{cont}_R : X \to Z$ is a flipping contraction i.e. a small birational morphism with $\rho(X/Z) = 1$ and $-(K_X + \Delta)$ ample over $Z$.

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- Note that after peturbing $\Delta$, we may assume that $\Delta \in \mathrm{Div}_{\mathbb{Q}}(X)$.

# Flips II

- The problem of existence of flips is local over $Z$ and hence we may assume that $Z$ is affine. It suffices then to show that $\mathfrak{R}(K_X + \Delta)$ is finitely generated. So this is a local version of the original problem.

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- To run this MMP with scaling, we must therefore show that such flips exist and terminate (i.e. there is no infinite sequence of such flips).
- Note that after each divisorial contraction, the Picard number drops by 1 and so there are only finitely many divisorial contractions.

# Outline of the talk

1. Introduction
2. Higher dimensional varieties
3. Towards the minimal model program in higher dimensions
   - Recent results
   - The structure of the proof
   - Existence of Flips
   - Existence of minimal models
   - Finiteness of models
   - Non-vanishing
   - Open problems

# Existence and termination of flips

**Theorem (Birkar-Cascini-Hacon-M$^c$Kernan)**

*Let $(X, \Delta)$ be a KLT log pair and $f : X \to Z$ a flipping contraction, then the flip $f^+ : X^+ \to Z$ of $f$ exists.*
*If moreover $\Delta$ is big, then any sequence of flips for the $K_X + \Delta$ MMP with scaling must terminate.*

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**Corollary**

*Let $(X, \Delta)$ be a KLT pair with $K_X + \Delta$-pseudo-effective and $\Delta$ big, then $K_X + \Delta$ has a minimal model. (In particular flips exist.)*

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$K_X + \Delta$-pseudo-effective means that given any ample $\mathbb{Q}$-divisor $H$, $K_X + \Delta + H$ is big (i.e. $\kappa(K_X + \Delta + H) = \dim X$). This is a necessary condition.

# Finite Generation

**Corollary (Birkar-Cascini-Hacon-M$^{c}$Kernan, Siu)**

*Let $X$ be a smooth projective variety, then $\mathfrak{R}(K_X)$ is finitely generated.*

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- As mentioned above, the general case is a consequence of a result of Fujino and Mori.
- Siu's proof relies on analytic techniques.

# The structure of the proof

We proceed by induction on $n = \dim X$. We show that for an $n$-dimensional KLT pair $(X, \Delta)$:

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All statements should be relative over a base.

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- The remaining implications heavily use the ideas of the MMP established by Kawamata, Kollár, Mori, Reid, Shokurov and others.

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- This is equivalent to showing that the algebra $\mathfrak{R}_S$ given by the image of the restriction map

$$\bigoplus_{m\in\mathbb{N}} H^0(\mathcal{O}_X(m(K_X+\Delta)))\to\bigoplus_{m\in\mathbb{N}} H^0(\mathcal{O}_S(m(K_S+\Delta_S)))$$

is finitely generated.

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- Roughly speaking we show that there exists a divisor $0 \leq \Theta \leq \Delta_S$ on $S$ such that

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- These extension theorems determine/define $\Theta$.

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- Proceeding this way we get a sequence of pairs $(X_i, \Delta_i + \lambda_i A_i)$ such that $K_{X_i} + \Delta_i + \lambda_i A_i$ is nef and trivial on some $K_{X_i} + \Delta_i$ extremal ray. We then flip/contract.

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- Each $(X_i, \Delta_i + \lambda_i A_i)$ is a minimal model for $(X, \Delta + \lambda_i A)$.

# Termination of MMP with scaling II

- If we could assume Claim $3_n$, we would be done. (As $\Delta$ is big, we may assume that $A' \leq \Delta \leq \Delta + tA$ for some ample divisor $A'$ and we let $\Delta_0 = \Delta + A$ .)

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- Since we can only assume $3_{n-1}$, we replace $(X, \Delta)$ by an appropriate DLT pair $(X', \Delta')$ and we run a MMP with scaling such that all flips/contractions are contained in $[\Delta']$.

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- Since we can only assume $3_{n-1}$, we replace $(X, \Delta)$ by an appropriate DLT pair $(X', \Delta')$ and we run a MMP with scaling such that all flips/contractions are contained in $[\Delta']$.
- This step is similar to Shokurov's reduction to PL-flips.

# Finiteness of models

## Theorem

*Assume that* $(X, \Delta_0)$ *is KLT,* $A$ *is an ample* $\mathbb{Q}$*-divisor and* $\Delta = \sum_{i=1}^{r} \delta_i \Delta_i$*. Let* $\mathcal{C} \subset [0,1]^r$ *be a rational polytope such that for any* $(\delta_1, \cdots, \delta_r) \in \mathcal{C}$*, the pair* $(X, \sum_{i=1}^{r} \delta_i \Delta_i + A)$ *is KLT. Then the set of isomorphism classes*

$$\{Y | Y \text{ is a min. model for } (X, \Delta + A);\ (\delta_1, \cdots, \delta_r) \in \mathcal{C}\}$$

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The proof is based on ideas of Shokurov (1996). It requires a compactness argument and so it is necessary to work with $\mathbb{R}$-divisors.

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- By the base-point free theorem, there is a morphism $f : X \to Z$ such that $K_X + \Delta \sim_{\mathbb{R}} f^*H$.
- Working over $Z$, we may assume that $K_X + \Delta \sim_{\mathbb{R}} 0$.
- Therefore $\mathcal{C} \cap U$ is a cone over $\Delta$ and we are done by induction on the dimension of the subspace of $\mathrm{Div}_{\mathbb{R}}(X)$ spanned by the components of $\Delta$.

# Non-vanishing

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- Using Kawamata-Viehweg vanishing, we can lift this section to $m(K_X + \Delta)$.

# Open problems

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- Existence of Minimal models for Log Canonical pairs (should be equivalent to Abundance).
- Termination of flips (would follow from the Borisov-Alexeev-Borisov conjecture).
- Characteristic $p$ (need resolution of singularities; there is no Kawamata-Viehweg vanishing Thm, but .... ).